IGCSE 0607 Practice Questions — E8.2 — Combined Questions & Markscheme

Vectors in two dimensions

Total Questions: 9 — **Total Marks:** 52 Marks

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- **E8.2** Vectors in two dimensions
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

(a) Work out

$$2\mathbf{p} + \mathbf{q}$$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(1, 4)$.
The translation from A to B is given by the vector $\mathbf{p}$.
Work out the coordinates of B.

(..... , ...) **[1]**

2 Given the vectors $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.

(a) Work out $3\mathbf{p}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

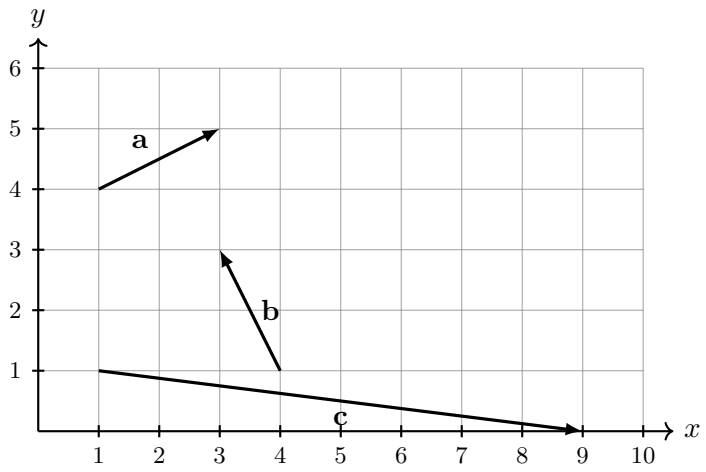
(b) Work out $\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point M is translated to point N by the vector $\mathbf{q}$.
Given that the coordinates of N are $(6, -1)$, find the coordinates of M.

(..... , ...) **[2]**

3 The diagram shows vectors **a**, **b** and **c** plotted on a Cartesian coordinate grid.



(a) Write **c** as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Vector **c** can be written in the form $p\mathbf{a} + q\mathbf{b}$, where p and q are integers.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[4]**

4 $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

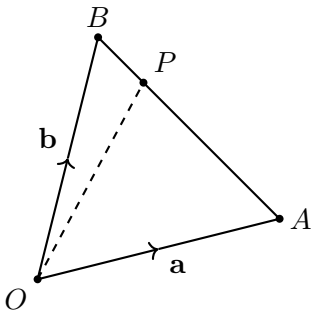
(a) Work out $3\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(3, -1)$.
$\vec{AB} = \mathbf{p} + \mathbf{q}$.
Find the coordinates of B.

(..... , ...) **[2]**

5 In triangle OAB, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
P is the point on AB such that $AP : PB = 3 : 1$.



(a) Find $\vec{OP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OP} =$ ..... **[3]**

(b) M is the midpoint of OA.
Find $\vec{MP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{MP} =$ ..... **[2]**

6 **(a)** $\mathbf{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$.

Given that $3\mathbf{u} + k\mathbf{v} = \mathbf{w}$, work out the value of k and the value of y.

$k =$ ..... , $y =$ ..... **[3]**

(b) Points P, Q and R have coordinates $(2, -1)$, $(5, 5)$ and $(11, 17)$ respectively.

(i) Write $\vec{PQ}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(ii) Show that P, Q and R lie on a straight line.

[2]

7 Points A, B and C have coordinates $(1, -2)$, $(7, 10)$ and $(4, 4)$ respectively.

(a) Write $\vec{AB}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Show that $\vec{AC} = \frac{1}{2}\vec{AB}$.

[2]

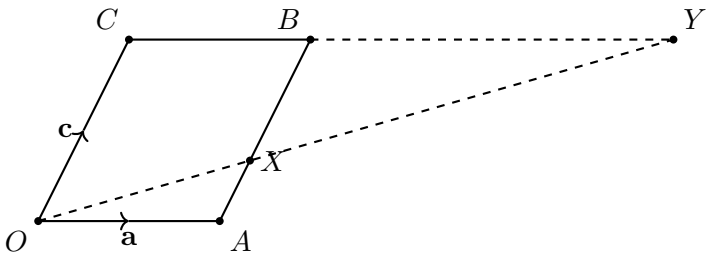
(c) Point C is translated by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$ to point D.
Find the coordinates of D.

(..... , ...) **[2]**

(d) Work out the vector $\vec{BD}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

8 $OABC$ is a parallelogram.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
X is a point on AB such that $AX : XB = 1 : 2$.
Y is a point on the line CB extended such that $CB : BY = 1 : 2$.



(a) Find $\vec{OX}$ in terms of **a** and **c**.

$\vec{OX} =$ ..... **[2]**

(b) Find $\vec{OY}$ in terms of **a** and **c**.

$\vec{OY} =$ ..... **[2]**

(c) Show that the points O, X and Y lie on a straight line.

[2]

(d) Find the ratio $OX : XY$.

$OX : XY =$ ..... **[1]**

9 Given the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

(a) Point P has coordinates $(5, -1)$.
The vector $\vec{PQ} = 2\mathbf{u} - \mathbf{v}$.
Find the coordinates of Q.

(..... , ...) **[3]**

(b) Point R has coordinates $(14, 3)$.
The vector $\vec{PR}$ can be written in the form $h\mathbf{u} + k\mathbf{v}$.
Work out the values of the constants h and k.

$h =$ ..... , $k =$ ..... **[5]**

1 **(a)** $\begin{pmatrix} 8 \\ -6 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ -1 \end{pmatrix}$ **[A1]**

(b) $(5, 1)$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $3 \times \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ or $\begin{pmatrix} 4 - 2(-2) \\ -3 - 2(5) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 8 \\ -13 \end{pmatrix}$ **[A1]**

(c) $\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ or $(6 - (-2), -1 - 5)$ **[M1]**

$(8, -6)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

3 **(a)** $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ **[B1]**

(b) Deduces correct column vectors for **a** and **b**: $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ **[B1]**

Sets up correct simultaneous equations from components: $2p - q = 8$ and $p + 2q = -1$ **[M1]**

Valid method to solve the simultaneous equations (e.g. multiplying first by 2 and adding: $4p - 2q = 16$

$5\text{p} = 15)[M1]$

$p = 3$ **[A1]**

$q = -2$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ or $\begin{pmatrix} -4 \\ 10 \end{pmatrix}$ seen **[M1]**

$\begin{pmatrix} 16 \\ -19 \end{pmatrix}$ **[A1]**

(b) $\vec{AB} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ **[M1]**

$(5, 1)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $\vec{AB} = \mathbf{b} - \mathbf{a}$ or $\vec{BA} = \mathbf{a} - \mathbf{b}$ **[B1]**

$\vec{OP} = \mathbf{a} + \frac{3}{4}(\mathbf{b} - \mathbf{a})$ or $\vec{OP} = \mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

(b) $\vec{MP} = -\frac{1}{2}\mathbf{a} + \vec{OP}$ (oe) **[M1]**

$-\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** Forms a correct equation for the horizontal components: $3(4) - k = 10$ **[M1]**

$k = 2$ **[A1]**

Forms a correct equation for the vertical components: $3(-3) + 2y = 6$

-9 + 2y = 6[M1]

$y = 7.5$ **[A1]**

(b) **(i)** $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[B1]**

(ii) Finds vector $\vec{QR} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}$ OR $\vec{PR} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}$ **[M1]**

States that $\vec{QR} = 2\vec{PQ}$ (or equivalent scalar multiplier), showing the vectors are parallel, AND mentions that they share a common point Q (or P), concluding they are collinear **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** $\begin{pmatrix} 7 \\ 10 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ 12 \end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix} 4 - 1 \\ 4 - (-2) \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[M1]**

$\frac{1}{2}\vec{AB} = \frac{1}{2}\begin{pmatrix} 6 \\ 12 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, hence $\vec{AC} = \frac{1}{2}\vec{AB}$ **[A1]**

(c) $(4 - 5, 4 + 2)$ **[M1]**

$(-1, 6)$ **[A1]**

(d) $\begin{pmatrix} -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 7 \\ 10 \end{pmatrix}$ or equivalent path method $-\vec{AC} + \vec{CD}$ **[M1]**

$\begin{pmatrix} -8 \\ -4 \end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

8 **(a)** $\vec{AX} = \frac{1}{3}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{3}\mathbf{c}$ **[A1]**

(b) $\vec{BY} = 2\mathbf{a}$ or $\vec{CY} = 3\mathbf{a}$ **[M1]**

$3\mathbf{a} + \mathbf{c}$ **[A1]**

(c) $\vec{OY} = 3\mathbf{a} + \mathbf{c} = 3(\mathbf{a} + \frac{1}{3}\mathbf{c}) = 3\vec{OX}$ **[M1]**

Since $\vec{OY}$ is a scalar multiple of $\vec{OX}$ and they share the common point O, they are collinear. **[A1]**

(d) $1 : 2$ **[B1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** $\vec{PQ} = 2\begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ -8 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ -8 \end{pmatrix}$ or $(5 + 7, -1 - 8)$ **[M1]**

$(12, -9)$ **[A1]**

(b) $\vec{PR} = \begin{pmatrix} 14 - 5 \\ 3 - (-1) \end{pmatrix} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$ **[B1]**

$3h - k = 9$ and $-2h + 4k = 4$ **[M1]**

Rearranging to get $k = 3h - 9$ and substituting: $-2h + 4(3h - 9) = 4$ or equivalent valid method to solve simultaneous equations **[M1]**

$10h = 40 \implies h = 4$ **[A1]**

$k = 3(4) - 9 = 3$ **[A1]**

[Back to Question] *[Total Marks: 8]*